Exercises week 1

8CM10 - Introduction to Quantum Physics and Structure of Matter

Special Relativity

1. The distance from New York to Los Angeles is about 5000 km and should take about 50 h in a car driving at 100 km/h.
    a. How much shorter than 5000 km is the distance according to the car travelers?
    b. How much less than 50 h do they age during the trip?

2. How fast must an object move before its length appears to be contracted to one-half its proper length?

3. For what velocity v is the total energy of a (relativistic) particle with mass m equal to twice its rest energy?

4. According to observer O, a blue flash occurs at $x_b = 10.4\ m$ when $t_b = 0.124\ \mu s$, and a red flash occurs at $x_r = 23.6\ m$ when $t_r = 0.138\ \mu s$. According to observer O', who is in motion relative to O at velocity u, the two flashes appear to be simultaneous. Find the velocity u.

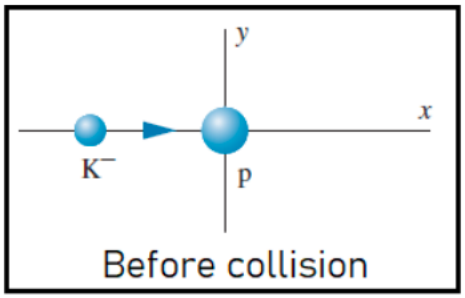
5. Two space exploration rockets approach the earth from opposing directions (say from x and $-x$). From the frame of earth, each of the rockets has a speed of $0.4c$. In the frame of rocket 1, its length is $30\ m$. What length does the rocket have in the frame of rocket 2?

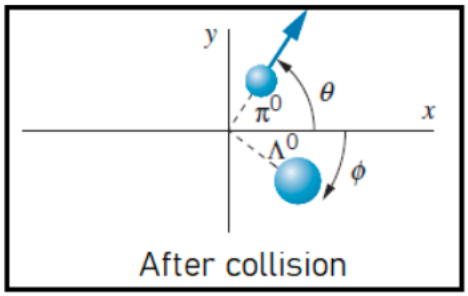
Relativistic Dynamics

6. The work-energy theorem relates the change in kinetic energy of a particle to the work done on it by an external force: $\Delta K = W = \int F dx$. Writing Newton's second law as $F = dp/dt$, show that $W = \int v\ dp$ and integrate by parts using the relativistic momentum to obtain:

$$K = \frac{mc^2}{\sqrt{1 - v^2/c^2}} - mc^2$$

7. Consider the reaction $K^- + p \rightarrow \Lambda^0 + \pi^0$ pictured below:





The rest masses and total energies are tabulated below:

Particle	Rest Masses (MeV/c^2)	Total energy (MeV)
K^-	493.7	646.1
p	983.3	983.3
Λ_0	1115.7	1194.6
π_0	135.0	389.8

Calculate the final momenta of π_0 and Λ_0, including their angles with respect to the xy-plane (θ and ϕ as defined in the picture).

8. An electron is moving at a speed of $0.81c$. By how much must its kinetic energy increase to raise its speed to $0.91c$?

9. What is the change in mass when $1g$ of copper is heated from 0 to $100°C$? The specific heat capacity of copper is $0.40\ J/g \cdot K$.

Photoelectric Effect

10. The work function ϕ (=energy to release one electron) for the photoelectric effect in potassium is 2.25 eV. For the case when light with a wavelength of 360 nm falls on the potassium, calculate
    a. The stopping potential V_s
    b. The kinetic energy and the velocity of the fastest of the emitted electrons

11. A homogeneous monochromatic light beam, with wavelength $400\ \text{nm}$, falls perpendicularly on a material with a work function of $\phi = 2.0\ \text{eV}$. The beam intensity is $3.0 \cdot 10^9\ \text{W/m}^2$. For the case when light with a wavelength of $360\ \text{nm}$ falls on the potassium.
    a. Assuming that each photon releases one electron, compute the number of electrons emitted per m^2 per second.
    b. Calculate the total absorbed energy
    c. Calculate the kinetic energy of the photoelectrons

12. On a graph of K_{max} versus f for the photoelectric effect, what is the physical significance of
    a. The slope
    b. The intercept with the horizontal axis f
    c. The intercept with the vertical axis K_{max}

Thermal Radiation

13. What is the temperature of a black sphere with a diameter of 10 cm which is emitting a total of 100 W thermal radiation? How much mass is lost each year by being radiated away? (use $E = mc^2$ to relate the mass loss with the emitted energy)

14. Assuming that the sun is a spherical black body with a radius $r = 7 \cdot 10^8$ m. The intensity I of the solar radiation at the surface of the earth (which is $1.5 \cdot 10^{11}$ m from the sun) is $1.4 \cdot 10^3$ W/m^2. Calculate the temperature of the surface of the sun.

15. The radiation law of Rayleigh Jeans says that $I(\lambda) = \frac{2\pi c}{\lambda^4} kT$, while Planck's radiation law is $I(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$. Show that for large wavelengths ($\lambda \to \infty$) the radiancy given by Planck's law approximates the Rayleigh Jeans radiancy.
    Hint: for small x it holds that $e^x \approx 1 + x$

16. The law of Stefan-Boltzmann for the total intensity *I* emitted by the surface of a black body with temperature T says: $I = \sigma T^4$. This law is consequence of Planck's radiation law: integrating the radiancy $R(\lambda)$ given by Planck's radiation law over all wavelengths λ yields the total intensity I.
    Derive the law of Stefan-Boltzmann and give the formula for the constant σ.
    Compute the numerical value of σ.
    Hint: put in the integral $x = \frac{hc}{\lambda kT}$ and use that $\int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}$.